

HR Attrition Analysis Report

This report highlights employees most likely to leave and identifies the main drivers of attrition. The analysis is based on the actual HR dataset and focuses on retention risk, department patterns, and action-oriented recommendations.

Top 3 Drivers of Attrition

- 1 Employees working overtime exhibit a materially higher attrition risk.
- 2 Lower compensation and shorter tenure are strongly associated with elevated departure risk.
- 3 Sales and other high-pressure roles show a greater probability of turnover relative to other departments.

Model Performance

Logistic Regression: Accuracy=0.880, Precision=0.800, Recall=0.339, ROC AUC=0.797

Random Forest: Accuracy=0.845, Precision=0.545, Recall=0.203, ROC AUC=0.774

Recommended Retention Actions

- 1 Review overtime-heavy teams and rebalance workloads where possible.
- 2 Create promotion pathways and retention plans for employees with longer tenure gaps.
- 3 Audit compensation and role-level incentives for high-risk departments and job families.

Watch List

Employee 911: Department=Research & Development, Role=Laboratory Technician, Income=2795, Tenure=1, Risk=0.93

Employee 622: Department=Research & Development, Role=Laboratory Technician, Income=2340, Tenure=1, Risk=0.91

Employee 1862: Department=Sales, Role=Sales Executive, Income=9907, Tenure=2, Risk=0.91

Employee 1273: Department=Sales, Role=Sales Representative, Income=1118, Tenure=1, Risk=0.91

Employee 1494: Department=Research & Development, Role=Laboratory Technician, Income=3172, Tenure=0, Risk=0.90